

PRACTICE PROGRAMS JAVA (6/8/26)

1.

Q1 Student

10 marks

Write a program to enter the mark of a student in four subjects. Then calculate the total and aggregate. Display the grade obtained by the student, if student scores an aggregate greater than 75% then grade is distinction

Your Code

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String []args){
4         Scanner sc = new Scanner(System.in);
5         int m1=sc.nextInt();
6         int m2=sc.nextInt();
7         int m3=sc.nextInt();
8         int m4=sc.nextInt();
9         int total = m1+m2+m3+m4;
10        double aggregate = total/4.0;
11        System.out.println("Total=" +total);
12        System.out.println("Aggregate = "+aggregate);
13        if (aggregate>75)
14            System.out.println("Grade = Distinction");
15        else
16            System.out.println("Grade = No Distinction");
17        sc.close();
18    }
19 }
```

Input

80
90
85
95

Output

Total=350
Aggregate = 87.5
Grade = Distinction

Run

Save

✓ Draft saved

2.

Q2 ATM

10 marks

write a program to print the total amount available in the atm machine with the conditions applied total denominations are 2000, 500, 200,100, get the denomination priority from the user and the total number of notes from the user to display the total available balance to the user

Your Code

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[]args){
4         Scanner sc = new Scanner(System.in);
5         int denomination = sc.nextInt();
6         int notes = sc.nextInt();
7         int total = 0;
8         if(denomination == 2000||denomination==500||denomination==200||
9            total = denomination*notes;
10            System.out.println("Total Amount = "+total);
11        }else{
12            System.out.println("Invalid Denomination");
13        }
14        sc.close();
15    }
16 }
```

Input

2000
5

Output

Total Amount = 10000

Run

Save

3.

Q3 TAX

10 marks

Write a program to calculate tax given following condition :

If income is less than or equal to 150000 then no tax

If taxable income is 150001 – 300000 the charge 10% tax

If taxable income is 300001 – 500000 the charge 20% tax

If taxable income is above 500001 the charge 30% tax

Your Code

```
1 public class TaxCalculator {
2     public static void main(String[] args) {
3         double income = 450000; // Change this value to test different
4         double tax = calculateTax(income);
5         System.out.println("Tax for income " + income + " is: " + tax);
6     }
7
8     public static double calculateTax(double income) {
9         double tax = 0;
10
11         if (income <= 150000) {
12             tax = 0;
13         } else if (income <= 300000) {
14             tax = (income - 150000) * 0.10;
15         } else if (income <= 500000) {
16             tax = (150000 * 0.10) + (income - 300000) * 0.20;
17         } else {
18             tax = (150000 * 0.10) + (200000 * 0.20) + (income - 500000) * 0.30;
19         }
20
21         return tax;
22     }
23 }
```

Input

stdin input

Output

Tax for income 450000.0 is: 45000.0

4.

Q4 BONUS

10 marks

a bonus 5% salary is given to grade A workers and 10% bonus to Grade B , write a program to enter salary and grade of employee , if salary less than employee gets an extra 2% binus on salary. Calculate the bonus that has to be given to the employee and print the salary that the employee will get

Your Code

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         double salary = sc.nextDouble();
8         char grade = sc.next().charAt(0);
9
10        double bonus = 0;
11
12        if (grade == 'A' || grade == 'a') {
13            bonus = salary * 0.05;
14        } else if (grade == 'B' || grade == 'b') {
15            bonus = salary * 0.10;
16        }
17
18        if (salary < 50000) {
19            bonus += salary * 0.02;
20        }
21
22        System.out.println(salary + bonus);
23    }
24 }
25
```

Input

40000
A

Output

42800.0